Component Specifications

Automatic Touchless Water Dispenser Using Arduino Uno

The following components are used to detect the user's hand, control the water valve, provide status feedback, and supply power to the automatic dispenser system.

Component	Function	Typical specification / design consideration
Arduino Uno	Main controller	ATmega328P-based development board; operates at 5 V logic; executes the control program and interfaces with the sensor, LCD, and servos.
HC-SR04	Hand-distance detection	Ultrasonic distance sensor; commonly used over approximately 2–400 cm; requires trigger and echo signals; used here with a 15 cm detection threshold.
I2C LCD 16×2	User feedback	16 columns × 2 rows; I2C interface reduces wiring; displays standby, dispensing, and water-stopped messages.
SG90 Servo ×2	Valve actuation	Small 5 V positional servos; approximate rotation range up to 180°; programmed positions: 0° for closed and 90° for open.
18650 battery pack	Power source	Rechargeable lithium-ion cell pack; must use a suitable battery holder, protection circuit, and charger; voltage and capacity depend on the number and arrangement of cells.
Power regulation	Provides appropriate supply	Regulates the battery voltage to suitable rails for the Arduino, sensor, LCD, and servos. The regulator must support the peak servo current.
Mechanical valve/tap	Controls water flow	Mechanical linkage operated by the two servos; should provide smooth movement, reliable sealing, and compatibility with the water line.

Power and Safety Considerations

• Use a common ground between the Arduino, sensor, LCD, servo supply, and control electronics.

• Do not power two SG90 servos directly from the Arduino 5 V pin if the supply cannot provide their combined peak current; a separate regulated servo supply is recommended.

• An 18650 battery pack requires proper polarity protection, overcharge protection, over-discharge protection, and a charger designed for the selected cell configuration.

• Keep electrical components protected from water and provide insulation between the water path and the electronics.

• Choose the regulator output voltage and current rating according to the actual Arduino, LCD, sensor, and servo requirements.

Component Interaction

The HC-SR04 sends hand-distance information to the Arduino Uno. When the measured distance is 15 cm or less, the Arduino updates the I2C LCD and commands both SG90 servos to actuate the mechanical valve/tap. The power-regulation stage converts the 18650 battery-pack output into stable supplies for the electronics and actuators.